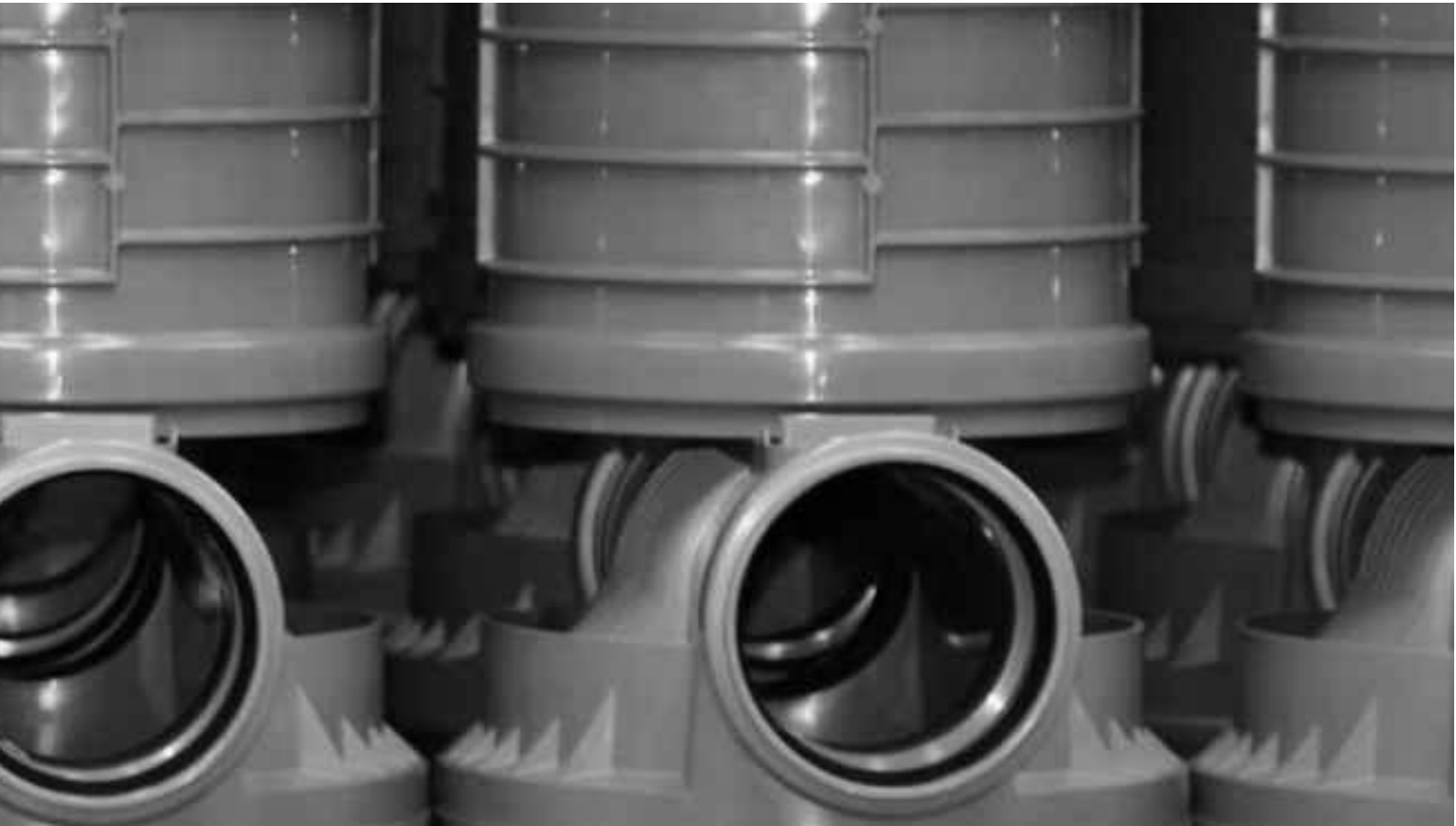


SMS Sewer Maintenance Structures



SMS/AXEDO DN375 MAINTENANCE SHAFTS

SMS/AXEDO DN375 MAINTENANCE SHAFTS

OVERVIEW



The SMS Maintenance Shafts are designed for sewer applications with strict quality compliance requirements, for example in terms of density and durability. Furthermore, these systems are designed to allow safe and time-efficient installation.

- We offer a range of Maintenance Shaft systems for the optimal collection of wastewater;
- Easy of access for maintenance equipment for pipeline systems (camera's, mirror & cleaning tools) which are permitted underground for inspection and maintenance;
- SMS – Maintenance Shafts provide alternate solution to traditional inspection shafts;
- SMS Maintenance Shafts can be installed below roadways and footpaths to an installation depth of 4m (max);
- Easily integrated to sewer networks configuration with diameter of DN100, DN150 & DN225;
- Designed, Engineered and Manufactured in Germany.

BENEFITS

- No Contamination of Soil and Groundwater;
- No Infiltration of pure water which could affect the operation of the water treatment system;
- SMS Maintenance Shaft guarantee density as synthetic pipeline systems and are conducted in accordance with the test standards;
- Easy manual installation as each element is no more than 25kg;
- Reduction of mechanical equipment with quicker installation over traditional shafts;
- Flat bottom of the SMS Maintenance Shafts provides a controlled stable platform;
- Coupling sleeves allow +/- 2 degrees for pipeline adjustments;
- SMS Maintenance Polypropylene shaft provides excellent resistance to various chemical agents;
 - External Sources – Aggressive Substrates;
 - Internal Sources – Transported Liquids such as Hydrogen Sulphide (H₂S), Sulphuric Acid (H₂SO₄) which is released by wastewater;
- SMS Maintenance Shafts are Resistance to Traffic Loads with specially developed top hat/access cover arrangements to facilitate distribution of the load requirements.

SPECIFICATIONS & REQUIREMENTS

Table 1 - SMS DN375 Maintenance Shaft (MS)

Riser Nominal Dia (DN)	DN375, DN300 or DN225
100% Recyclable	Yes
Person Entry	No
Chemical Resistance	pH 1-13
Base Material	Polypropylene (PP)
Reducing Cone Material	N/A
Riser Pipe Material	PVC Pipe
Gasket Material	Styrene Butadiene Rubber (SBR)
Max Installation Depth	4
Max Ground Water Height	5m above inflow pipe invert level
Joint Grade & Deflection Angle	±2°
Max Traffic Load	Class D to AS3966 "Access Covers and Grates"
Pipe Inlet & Outlet Material	Drainage Waster & Vent (DWV) to AS/NZ1260
Pipe Inlet & Outlet Diameter	DN100, DN150 or DN225
High Level Inlet Connection	Drop Tees
Product Certification	Globalmark - WSA137
WSAA Product Appraisal	PA 1317 – Issue 4
WSAA Product Specification	PS-341 Maintenance Shafts (MS) (PP) for Non-Pressure Applications - Gravity Sewerage
Relevant Standards	WSA137, EN13598-2, AS3996, AS1646
Service Life	100 Years

MAINTENANCE SHAFT BASE CONFIGURATIONS

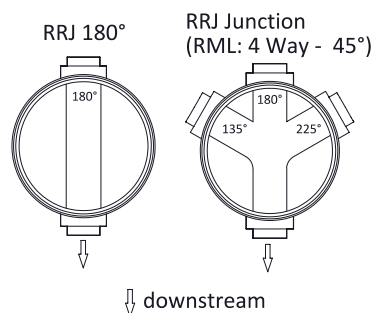


Table 2 - SMS DN375 Maintenance Shaft Configurations

Product Code	Configuration	Inlet/Outlet Size
MS8	DN375 - 180° Straight	DN150
MS9	DN375 - RML (4-Way)	DN150
MS14	DN375 - 180° Straight	DN225

SMS/AXEDO DN375 MAINTENANCE SHAFTS

TECHNICAL FEATURES

Chemical and Corrosion Resistance

Polypropylene materials provide excellent resistance to various chemical agents.

Polypropylene is particularly effective in withstanding potential deterioration from:

- External sources: due to aggressive substrates;
- Internal sources: due to transported liquids, such as hydrogen sulphide (H2S) and sulphuric acid (H2SO4) released by the wastewater.

Resistance to Traffic Loads

The specially developed top hats & access cover arrangement facilitates the distribution of the load, as a result of which the load is spread across the substrate. Follow all installation guides including minimum gap allowances to ensure any loads are not directly transferred to the riser pipe.

Accessibility

With an extensive range of options, the SMS Maintenance Shafts allow you to anticipate all possible configurations. Our specialists will assist you in choosing the right solution.

Ease of Inspection

The flow sections have a continuous longitudinal profile that facilitates entry for inspection and cleaning equipment.

Ease of Maintenance

The extremely low wall-roughness prevents the deposit and adherence of dirt. This makes maintenance of the SMS Maintenance Shaft faster than it is when traditional materials are used.

Recycling

Polypropylene is a thermoplastic material. It is therefore entirely recyclable just like all other thermoplastic materials used in sewer and pressure systems (PVC, PE).

Ease of Installation

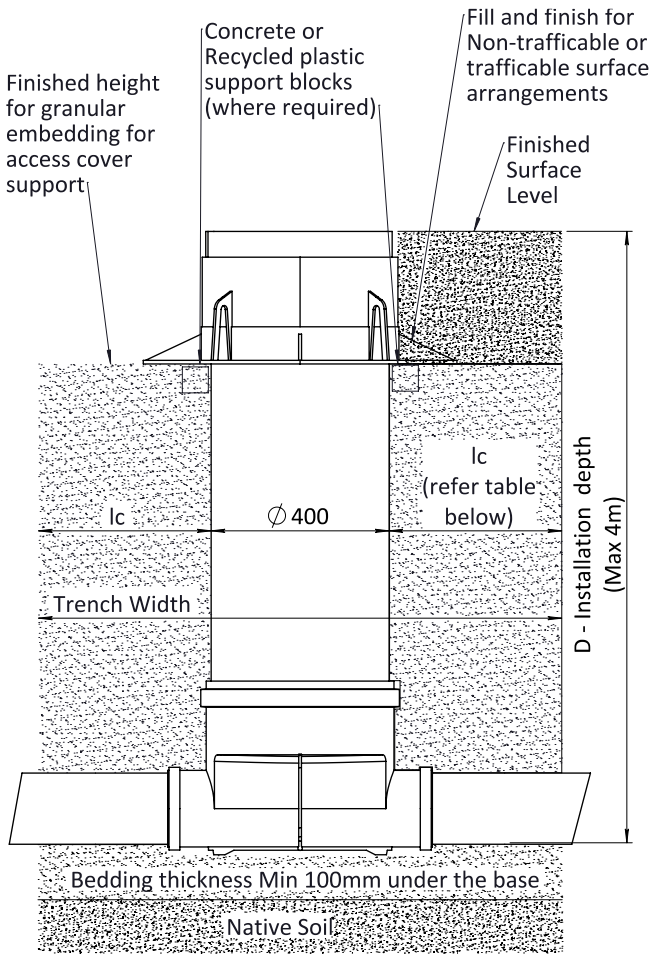
The SMS Maintenance Shaft facilitates easy manual installation (every element separately does not weigh more than 25 kg, apart from the access cover or concrete components). The installation does not require any mechanical equipment can be completed quicker than the installation of traditional maintenance shafts.

During installation (e.g. underground obstacles, cables and pipelines, etc.), it is possible to displace the inlet/outlet socket by an angle of $\pm 2^\circ$. A flat bottom provides for a stable SMS Maintenance Shaft. Contact SMS for detailed installation guides.

The products shown form part of SMS's continuous improvement program and as such the product designs, specifications and materials may change without notice. All warranties relating to accuracy, completeness, or suitability for any particular purpose and all liability for any loss, damage or costs incurred relating to the use of this information are excluded.

INSTALLATION OVERVIEW

General overview of DN375 Maintenance Shaft utilizing DN375 Top hat. For installation of different installation options, contact SMS for detailed installation guides.



Installation notes:

- Refer to local Water Agency specifications for other specific installation guidelines.
- For installation in reactive soils, refer to local water agency installation guidelines.
- Concrete used for installation or concrete components to comply to local water agency specifications.
- Test installed maintenance shaft to local water agency requirements.
- Refer to AS/NZS 2566 for Buried Flexible Pipelines and local authority requirements for any specific installation requirements.

Bedding, Backfilling/Embedment material:

- The bedding shall be granular material, embedment sand (WSA-PS-360) or 5mm minus crushed rock (WSA-PS-361).
- Compaction: DPr $\geq 97\%$.

Trench widths	Ic (Min.)	Trench Width (Min.)
Standard Installation	400	800
Groundwater conditions	500	1400

Installation depth notes:

- Standard Installation max 4m depth, in high water table/groundwater conditions max xm depth of installation.

Figure 1 – Typical Installation using Top Hat

MAINTENANCE SHAFT CONFIGURATION OPTIONS

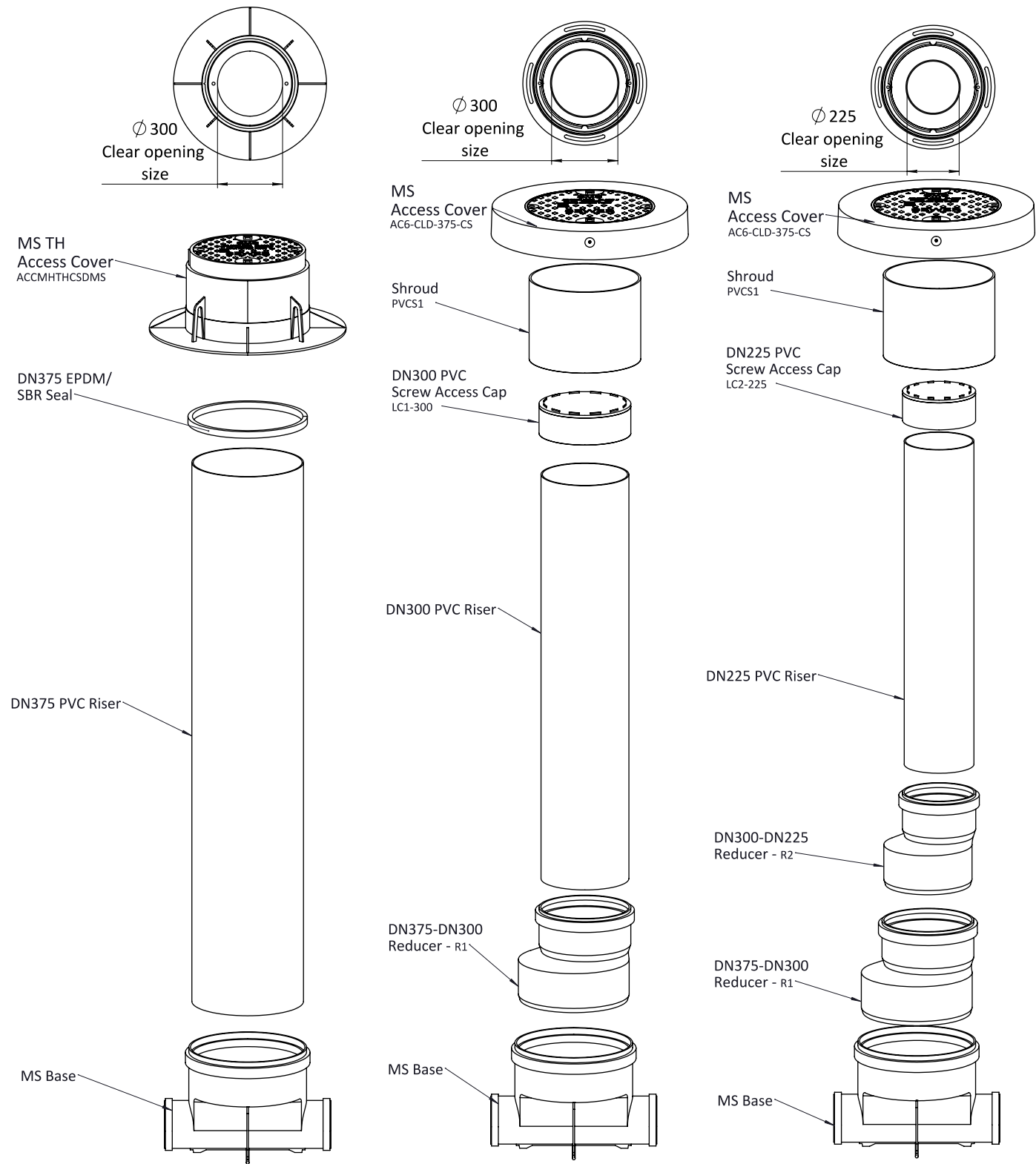


Figure 2 – Typical Installation options

SYSTEM COMPONENTS

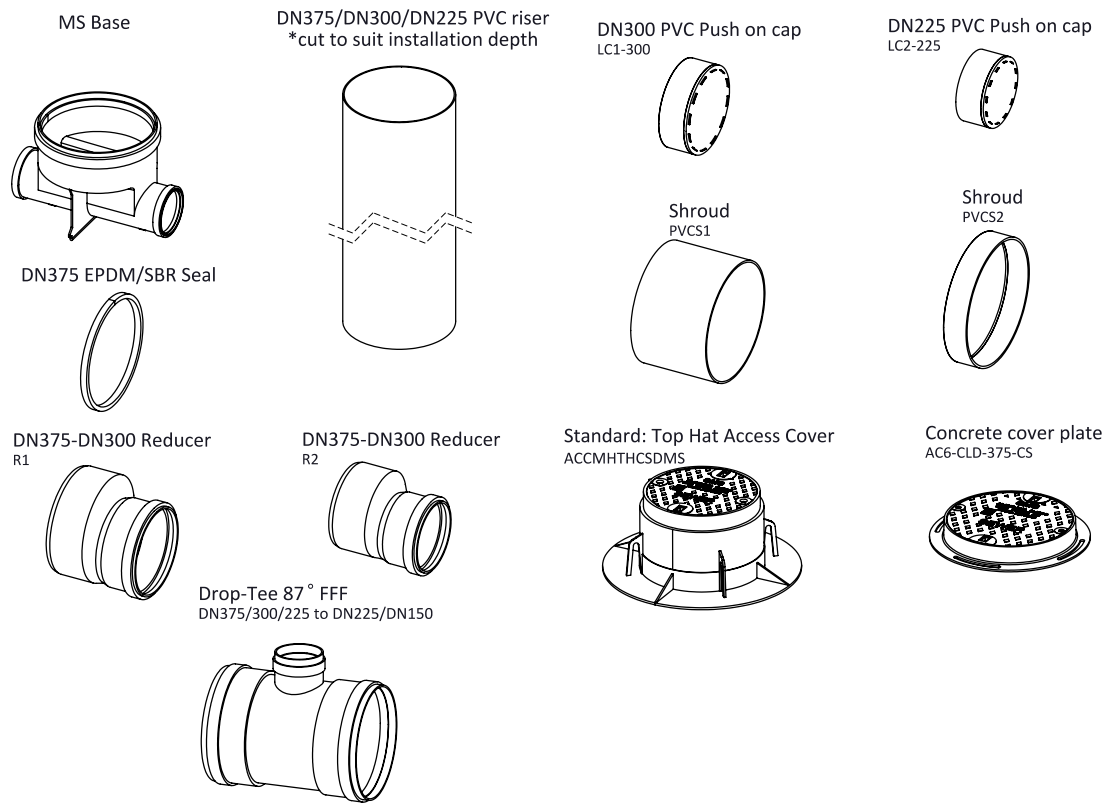


Figure 3 – Typical System Components

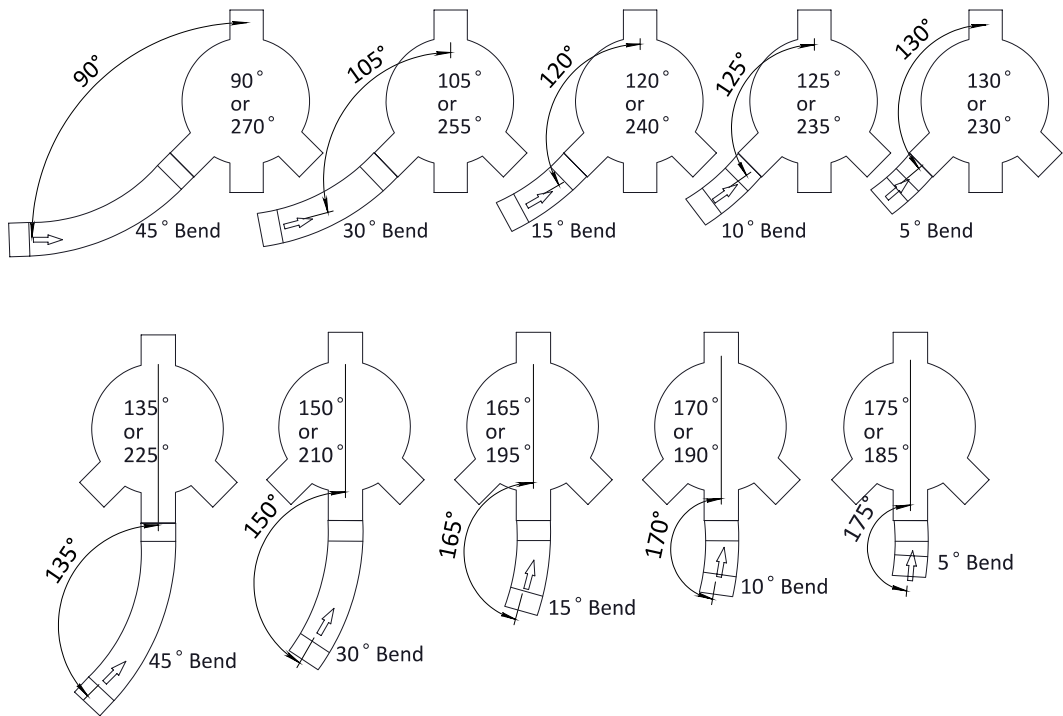


Figure 4 – Angles achieved using long radius bends

Contact SMS for additional configuration detail to meet project specifications.

CONTACT SMS

Sales, Product & Technical Information: 1300 482 228

Email: sales@sewerms.com.au

Website: www.sewerms.com.au

KEY CONTACTS

Gavin Roberts
Sales & Marketing Director – APAC
M 0450 204 334

Peter Fox
Technical, Sales Business Development Manager
M 0478 084 333

Alex Waites
Client Services East Coast
M 0419 853 361

Locations: Melbourne/Sydney/Brisbane/Townsville/Perth/Adelaide/Tasmania/Pacific Region

